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Subject Name & Code: Cyber Security (AD31234B)

Assignment No: 05

Title of Assignment: Intrusion Detection system with the help of appropriate tool and decide the design methodology for smart intrusion prevention system.

Date of Performance: /07/2025

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Aim: Study and demonstrate Intrusion Detection system with the help of appropriate tool and decide the design methodology for smart intrusion prevention system.

Objective:

1. discuss about why we require IDS and IPS system require?
2. Refer the documents attached
3. Search and study Snort tool
4. Download link for Snort tool <https://www.snort.org/#documents> or <https://www.snort.org/downloads>
5. Refer user manual & also both the write-up attached here for reference
6. Include download steps screenshots and demonstration of commands snapshots in write-up

Materials Required:

Computer with internet access
Snort Tool installed in the machine

Procedure:

1. Introduction

With the rapid growth of the internet and interconnected devices, network security has become one of the most critical concerns. Traditional security mechanisms such as firewalls and antivirus software are often insufficient against sophisticated cyberattacks. This is where **Intrusion Detection Systems (IDS)** and **Intrusion Prevention Systems (IPS)** come into play.

An IDS monitors network traffic to detect malicious activity, while an IPS goes one step further by actively blocking or preventing those intrusions in real time.

One of the most widely used IDS/IPS tools in both academia and industry is **Snort**, an open-source intrusion detection and prevention system developed by Cisco.

2. Why Do We Require IDS and IPS?

Intrusion Detection System (IDS)

- **Definition:** IDS is a security mechanism that monitors network or system activities for malicious actions or policy violations. It works as an alerting system, notifying administrators of potential threats.
- **Need for IDS:**
 - Detect unauthorized access attempts.
 - Identify malware, worms, or insider threats.
 - Monitor suspicious traffic patterns and generate alerts.
 - Provide forensic data for post-attack analysis.
- **Limitation:** IDS is passive – it only detects and alerts but does not stop the attack.

Intrusion Prevention System (IPS)

- ❑ **Definition:** IPS extends IDS functionality by actively responding to detected intrusions, often by dropping malicious packets, blocking traffic, or reconfiguring firewalls.
- ❑ **Need for IPS:**
 - Prevent attacks in real time (instead of just alerting).
 - Protect against zero-day vulnerabilities.
 - Automatically mitigate denial-of-service (DoS) attacks.
 - Reduce human intervention by automating responses.

Aspect	IDS	IPS
Function	Monitors and alerts	Monitors + blocks/prevents
Nature	Passive	Active
Response	Logs, notifications	Drops packets, blocks traffic
Placement	Outside traffic flow (monitoring port)	Inline, between firewall and network
Use Case	Forensics, auditing	Real-time protection

Note: In practice, modern security devices often combine IDS and IPS into a single **IDPS system**.

3. Introduction to Snort

Snort is the most popular **open-source IDS/IPS tool**. It was initially released in 1998 and is now maintained by Cisco. Snort can analyze network traffic in real time and detect various types of attacks, including buffer overflows, stealth port scans, CGI attacks, SMB probes, and more.

Modes of Operation in Snort

1. **Sniffer Mode:** Captures and displays packets on the network.
2. **Packet Logger Mode:** Logs packets to disk for later analysis.
3. **Network Intrusion Detection System (NIDS) Mode:** Uses rules/signatures to detect and alert suspicious traffic.
4. **Inline Mode (IPS):** Drops or rejects packets in real time if they match malicious signatures.

Snort Architecture

- ❑ **Packet Decoder:** Reads raw packets from network.
- ❑ **Preprocessors:** Normalize and preprocess traffic (e.g., HTTP decoding).
- ❑ **Detection Engine:** Applies rules/signatures to packets.
- ❑ **Logging and Alerting System:** Logs data or sends alerts.
- ❑ **Output Modules:** Exports results to files, databases, or consoles.

Features of Snort

- ❑ Open-source and free (community version).
- ❑ Real-time traffic analysis and packet logging.
- ❑ Protocol analysis (IP, TCP, UDP, ICMP).
- ❑ Content searching and matching.
- ❑ Supports rule-based language for defining attacks.
- ❑ Can act as IDS (alert only) or IPS (block traffic).
- ❑ Integrates with SIEMs (Security Information and Event Management)

SNORT IMAGES AND STEPS OF INSTALLATION

5. Design Methodology for Smart Intrusion Prevention System

A **Smart IPS** should go beyond traditional rule/signature-based detection. The design methodology includes:

1. **Hybrid Detection Approach:**
 - o **Signature-based detection** for known attacks.
 - o **Anomaly-based detection** using machine learning for unknown/zero-day attacks.
2. **Real-Time Blocking:**
 - o Inline deployment for dropping malicious packets immediately.
3. **Adaptive Learning:**
 - o System should learn from historical attack data.
 - o Use AI/ML to improve detection accuracy.
4. **Low False Positives:**
 - o Optimize rules to prevent unnecessary blocking of legitimate traffic.
5. **Integration with SIEM/Firewalls:**
 - o IPS should communicate with firewalls, log servers, and monitoring tools.
6. **Scalability & Performance:**
 - o Must handle high-speed gigabit networks without delay.

Ex: A smart IPS would use Snort's rule-based detection combined with a machine learning model trained on network traffic patterns to identify and block anomalies in real time.

Conclusion:

Intrusion Detection and Prevention Systems are essential components of modern cybersecurity architecture. While IDS like Snort can detect attacks and alert administrators, IPS extends this capability by actively preventing intrusions in real time.

Snort, being open-source and highly flexible, provides an excellent platform to study IDS/IPS functionalities. By demonstrating Snort in different modes, we can understand how packets are captured, logged, analyzed, and blocked.